

ANIMESH GIRI

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SUMMARY

Data Scientist and AI/ML Engineer with 3+ years of experience building machine learning pipelines, predictive models, and LLM-driven applications across healthcare, enterprise, and research settings. Experienced in translating complex structured and multimodal data into production-ready solutions that improve decision-making, automation, and operational efficiency. Strong background in Python, SQL, deep learning, time-series modeling, ETL development, and cloud-based data systems.

PROFESSIONAL EXPERIENCE

Research Assistant II, Data Science and AI

Brigham and Women's Hospital | Boston, MA | Aug 2025 - Feb 2026

- Built MATLAB batch processing pipelines for multi-channel neonatal physiological data, reducing preprocessing time by 67 percent and accelerating research model iteration.
- Developed an autoencoder-based artifact detection system that improved pulse-to-noise ratio by more than 30 percent, increasing input stability for downstream blood pressure prediction models.
- Evaluated CNN-LSTM architectures for continuous blood pressure prediction, analyzing model error patterns and failure modes to support sensor configuration decisions.
- Designed and validated an LLM-based grading assistant that achieved 90 percent numeric scoring accuracy and 100 percent divisional agreement, reducing clinician review time and improving consistency.

AI Engineer, Project Based

Humanitarians AI | Boston, MA | Jan 2025 - Jun 2025

- Architected multi-agent LLM workflows using OpenAI and Claude APIs with structured prompt pipelines, improving reasoning consistency across multi-step analytical tasks.
- Built a Streamlit evaluation dashboard to monitor agent outputs, prompt performance, and iteration metrics, enabling rapid prototyping and stakeholder demonstrations.

R&D Advanced Intern, Data Science

Tides Medical | Wayland, MA | Jun 2024 - Dec 2024

- Designed Python ETL workflows with schema validation and integrity checks, improving processing efficiency by 40 percent and enabling more reliable analytics.
- Automated statistical testing workflows including t-tests and Kruskal-Wallis analysis, integrating outputs into interactive dashboards and reducing manual analysis time.

Junior Data Scientist

Indium | India | Jan 2021 - Aug 2022

- Engineered advanced SQL transformations using CTEs and window functions on SAP ERP datasets, reducing reporting latency from 12 hours to near real-time.
- Built business intelligence dashboards in Power BI for operational and client reporting, supporting KPI tracking and decision-making.
- Developed churn prediction and TF-IDF recommendation models that improved retention accuracy by 15 percent and increased digital engagement by 20 percent.
- Built ARIMA and XGBoost forecasting models that improved demand planning accuracy by 15 percent and supported inventory optimization.

TECHNICAL SKILLS

Programming and Databases: Python, R, SQL, PostgreSQL, SQL Server, Snowflake

Machine Learning and AI: Scikit-learn, PyTorch, TensorFlow, supervised learning, unsupervised learning, deep learning, CNNs, RNNs, transformers, NLP, LLMs, prompt engineering, feature engineering, SHAP, predictive modeling, statistical modeling, exploratory data analysis, time-series forecasting, A/B testing

Data Engineering: ETL pipelines, REST APIs, Apache Airflow, Spark, dbt, SSIS, Talend, Alteryx, data warehousing

Cloud and DevOps: Azure Data Factory, Azure SQL, Azure AI Foundry, Microsoft Fabric, AWS S3, AWS Glue, AWS Lambda, Docker, Kubernetes, CI/CD, Git

Enterprise and Analytics Tools: SAP ERP, Salesforce, ServiceNow, JIRA, Power BI, Tableau, Excel, JMP, Matplotlib, Seaborn

PROJECTS

Supply Chain Demand Forecasting and Inventory Optimization

- Built an end-to-end forecasting pipeline on the Walmart M5 dataset using SKU-level feature engineering and XGBoost.
- Developed a Streamlit decision-support application for inventory risk assessment and KPI monitoring.

Stroke Prediction Model

- Developed CNN and ensemble machine learning models for stroke prediction, achieving 95 percent accuracy.
- Applied SMOTE to address class imbalance and improved minority-class sensitivity by 25 percent.

BookBot, Book Recommendation System

- Built semantic search across more than 140,000 book summaries using SBERT embeddings and Pinecone.
- Integrated a GPT-based conversational interface to improve recommendation relevance and retrieval precision.

EDUCATION

Northeastern University | Boston, MA

M.S. in Information Systems, Data Science Focus | Sep 2022 - May 2024

University of Mumbai | India

B.E. in Electronics and Telecommunication Engineering | Aug 2018 - May 2022

CERTIFICATIONS

- Microsoft Azure AI900: Azure AI Fundamentals
- Microsoft MTA:98-381 Introduction to Python Programming